

Assignment 1
Dateline: 2/10/2025
Variables, Methods, Conditionals, Loop

Use Dr Java to answer two programming challenges. Only technical strategies covered in ZZ1102 lectures are allowed to solve all problems. Answer all and submit your **ONLY ONE** zip file of all your .java files on Canvas.

Note: Kindly include comments (documentation) in your codes, to explain what each statements do.

Strictly no copying/help from each other. The code will be checked for plagiarism/copying through a Software so be careful. In case of copying/plagiarism, the report generated by the Software will be shared with you to give you a proof of copying.

Challenge 1 [60 marks]

You are required to develop a 10 level tower game. You need a class named TowerGame to implement the program. The descriptions of class are as follows:

Implement a 10 level tower game. The game is played between 2 players. The game goes as follows:

- Each player takes turn to draw two random numbers.
- If the addition of these two numbers is an even number, the player may climb one level up the tower. If the addition of these two numbers is an odd number, the player fall one level down the tower.
- Any player who gets to the top of the tower first will be the winner.

You are required to implement the necessary methods in your implementation.

Challenge 2 [60 marks]

You are required to develop a computer game. You need a class named TankGame to implement the program. There are 3 small tanks and one MainTank. The objective of the game is to collect balls from each small tank into the MainTank. The descriptions of the game are as follows:



1. The game should start with random number of balls (minimum 0 and maximum 5) in each small tank.
2. Each ball can only travel from one small tank to another in order to reach the mainTank.
3. The game will terminate once all the balls from each tank moved into the MainTank.



Implement the game. You are required to at least include the following:

- 1) To use void and value methods to implement the strategies in the game.
- 2) To implement the main method